Wind Field Generation

Exercise to Lecture #7
Controller Design for Wind Turbines and Wind Farms

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1 Wind Field Generation

Here, we only consider one point at a hub height of 90 m.

a) Calculate the amplitudes of an IEC Kaimal longitudinal spectrum [1] at 20 m/s and turbulence class A for the following frequencies: 0.001, 0.01, 0.1, and 1 Hz. Use 0.001 Hz for the frequency resolution/discretization.
b) Generate a wind time series for IEC Kaimal longitudinal spectrum [1] at 20 m/s and turbulence class A, with a length of $T = 4096$ s and a time discretization of $\Delta t = 0.25$ s considering frequencies up to $f = 2$ Hz. Use the seed 1 and uniformly distributed phase angles between 0 and 2π. Compare the wind speed generated by a sum over individual frequencies and the wind speed generated by `ifft`.
c) Show by numerical integration that the standard deviation of the time series and the square root of the area below the analytic spectrum are close to the value specified in the standard.
d) Compare the analytic spectrum with an estimated spectrum using `fft`.
e) Compare the analytic spectrum with an estimated one using `pwelch`. First, use a single block, a rectangular window, and no overlap. Then use at least 8 blocks, a Hamming window with the block length, and 50 % overlap. Use the second setting for the rest of the exercise.
f) Generate the analytic spectrum of the rotor-effective wind speed (grid resolution of 4 m) for a rotor diameter of 126 m and compare it to the single-point analytic spectrum.

2 Wind Field Analysis

a) Generate a turbulent wind field with the same grid resolution and spectral properties as above using TurbSim.
b) Read the wind field into MATLAB with readBLgrid.m.
c) Compare the analytic spectrum of a single point with an estimated one for the wind at hub height using `pwelch`.
d) Compare the analytic spectrum of the rotor-effective wind speed with the estimated one using `pwelch`.
e) Compare the magnitude-squared coherence from [1] (so you have to square "Coh(r,f)") to the estimated one from two points at hub height with a distance of 4 m using `mscohere` with at least 16 blocks. For more details about the bias in coherence calculations see [2].

References

[1] IEC 61400-1. *Wind turbines - Part 1: Design requirements*. International Electrotechnical Commission, 2005.

[2] J. S. Bendat and A. G. Piersol. *Random data; analysis and measurement procedures*. New York, USA: John Wiley & Sons, 1971.